



Beijing-Dublin International College



SEMESTER II FINAL EXAMINATION – 2017/2018

Beijing-Dublin International College
BDIC1032J Maths 4 (Advanced Mathematics; Engineering)

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MODULE LECTURERS: Yanru Ping, Yuehong Feng

Time Allowed: 90 minutes

Instructions for Candidates

Answer ALL questions. The marks that each question carry is written as shown.

BJUT Student ID: _____ **UCD Student ID:** _____

I have read and clearly understand the Examination Rules of both Beijing University of Technology and University College Dublin. I am aware of the Punishment for Violating the Rules of Beijing University of Technology and/or University College Dublin. I hereby promise to abide by the relevant rules and regulations by not giving or receiving any help during the exam. If caught violating the rules, I accept the punishment thereof.

Honesty Pledge: _____ **(Signature)**

Instructions for Invigilators

Non-programmable calculators are permitted. NO dictionaries are permitted.
No rough-work paper is to be provided for candidates.

NOTE: Answer **ALL** questions.

Time allowed is **90** minutes.

The exam paper has **2** sections on **5** pages, with a full score of 100 marks.

You are required to use only the provided **Examination Book** for answers.

SECTION A — GAP-FILLING QUESTIONS

This section is worth a total of **60** marks, with each question worth **5** marks.

1. Let $f(x, y)$ be a continuous function in the neighborhood of the origin $O(0, 0)$. Given

$$\lim_{x \rightarrow 0, y \rightarrow 0} \frac{f(x, y) - f(0, 0)}{x^2 + y^2} = 1,$$

which of the following statements is true? _____

- (a) Point $O(0, 0)$ is not a point of extreme value for $f(x, y)$;
- (b) Point $O(0, 0)$ is a point of maximum value for $f(x, y)$;
- (c) Point $O(0, 0)$ is a point of minimum value for $f(x, y)$;
- (d) It cannot be decided whether the Point $O(0, 0)$ is a point of extreme value for $f(x, y)$ with the given condition.

2. Determine the limit

$$\lim_{(x, y) \rightarrow (1, 1)} \frac{\sqrt{xy + 1} - 1}{x + y} = \underline{\hspace{2cm}}.$$

3. Given $z = \ln \sqrt{x^2 + y^2}$, find the total differential at the point $(1, 2)$

$$dz \Big|_{(1, 2)} = \underline{\hspace{2cm}}$$

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4. For a function $z = f(x, y)$, there is a statement:

If $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ exist at the point $P_0(x_0, y_0)$, then $z = f(x, y)$ is continuous at $P_0(x_0, y_0)$.

Determine this statement is true or false: _____

5. Let $z = z(x, y)$ be an implicit function given by the equation

$$F\left(\frac{x}{y}, \frac{z}{x}\right) = 0,$$

where $F(u, v)$ is derivable. Then

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = \underline{\hspace{2cm}}$$

6. Consider a function

$$z = f(2x - y, 3x + 2y),$$

where $f(u, v)$ has second order continuous partial derivatives. Then $\frac{\partial^2 z}{\partial x \partial y}$ is _____.

7. Let Σ be a surface described by the equation

$$z = \arctan \frac{y}{x}.$$

The equation of the tangent plane to Σ at the point $P\left(1, 1, \frac{\pi}{4}\right)$ is: _____.

And the equation of the normal line is given by _____.

8. The gradient of the function

$$z = 3 + xy - x^3 - y^3$$

at the point $P(-1, 1)$ is _____.

And what is the largest directional derivative at $P(-1, 1)$? _____

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9. Find the points on the sphere $x^2 + y^2 + z^2 = 4$ that are closest to and farthest from the point $P(3, 1, -1)$, by making use of the Lagrange multiplier method.

(a) Write down the auxiliary function: _____.

(b) Find the required points.

10. Given a function

$$z = (2x^2 - 4x + y) e^y,$$

find its extreme values.

11. Calculate the double integral

$$I = \iint_D x [1 + y f(x^2 + y^2)] dx dy, \quad f(u) \text{ — continuous over } D,$$

where D is the region bounded by $y = x^3, y = 1, x = -1$.

12. Suppose $f(x, y)$ is a continuous function over a region D , satisfying

$$f(x, y) = xy + \iint_D f(x, y) dx dy,$$

where D is enclosed by

$$y = 0, \quad y = x^2, \quad x = 1.$$

Then $f(x, y) =$ _____

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SECTION B — EXTENDED ANSWER QUESTIONS

Write your answers on the **Examination Book** provided.

This section is worth a total of **40** marks. The marks of each question are as shown.

- 13. (10 marks)** Compute the following double integral by converting it to the polar coordinates

$$I = \iint_{D: \quad x^2+y^2 \leq 4, \quad (x+1)^2+y^2 \geq 1} \left(\sqrt{x^2+y^2} + y + 1 \right) dx dy.$$

**Hint:* First, sketch the region D . Second, you can use the property of symmetry to simplify your operation, and then express the double integral as an iterated integral in the form of polar coordinates. Last, find the value of the double integral.

- 14. (7 marks)** Evaluate the following triple integral by applying cylindrical coordinate system

$$I = \iiint_{\Omega} (x^2 + y^2) dv,$$

where Ω is a region enclosed by the plane $z = 8$ and the rotating paraboloid $2z = x^2 + y^2$.

- 15. (7 marks)** Calculate the double integral

$$I = \iint_D e^{\max\{x^2, y^2\}} dx dy, \quad \text{where } D = \{(x, y) \mid 0 \leq x \leq 1, \quad 0 \leq y \leq 1\}.$$

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- 16. (8 marks)** Express the following triple integral as an iterated integral in spherical coordinates:

$$I = \iiint_{\Omega} f(x, y, z) \, dv,$$

where Ω is a region bounded by

$$\text{the cone } z = \sqrt{3(x^2 + y^2)} \quad \text{and the hemisphere } z = \sqrt{4 - x^2 - y^2}.$$

- 17. (8 marks)** Let Ω be a 3 dimensional solid body bounded by the surfaces

$$z = 2x^2 + y^2 \quad \text{and} \quad z = 6 - (x^2 + 2y^2).$$

Compute the volume of Ω .

Glossary

Additivity	可加性
Auxiliary function	辅助函数
Cartesian form	笛卡尔形式
Cone	锥面
Cylinder	柱面
Cylindrical coordinate	柱坐标
Derivable	可导的
Directional derivative	方向导数
Ellipsoid	椭球面
Extreme value	极值
Gradient	梯度
Iterated integral	累次积分
Lagrange multiplier method	拉格朗日乘数法
Maximum value	极大值
Minimum value	极小值
Normal line	法线
Paraboloid	抛物面
Polar coordinate	极坐标
Sphere	球面
Spherical coordinate	球坐标
Symmetry	对称性
Tangent plane	切平面
Total differential	全微分